Northstar Build Document

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1. Overview

Northstar is a multi-tenant WhatsApp-first business OS that takes a lead from first message to paid, scheduled and completed work with AI and automation. Every lead source opens a WhatsApp chat; the AI qualifies and scores the lead, sends a quote, collects payment through Razorpay, issues a GST invoice, books free staff from their Google Calendars, collects the balance, and feeds conversions back to Meta so the next ads bring better leads.

Who it is for: small and mid-size Indian service businesses that sell through WhatsApp and ads: shoot production, agencies, home services, coaching and similar.

First customer: Reelrr (shoot production). Reelrr is the reference workflow, not the product boundary; the platform is configurable for any industry.

Product goals

- One record per customer from first message to final payment
- Hands-free lead handling, with staff approval where the business wants it
- Ads optimised on real buyers, not just chats
- Live visibility in a dashboard and Google Sheets

Domains: `northstar.com` (marketing site), `app.northstar.com` (the platform), `northstar.com/c/<code>` (tracked chat links).

2. Scope

The MVP covers the full lead-to-payment loop for any business on WhatsApp; visual workflow building, voice calls and extra lead platforms come later.

Area	MVP (pilot)	Phase 2
Lead sources	WhatsApp only: Meta click-to-WhatsApp ads, organic WhatsApp, tracked chat links and QR codes for website, email, social media	IndiaMART, voice calls, direct website forms
Workflow	Configurable stages, step on/off, 4 industry templates + blank, simple conditions, approval mode toggle	Visual drag-and-drop builder, complex branching
AI	Qualifying conversation, extraction, rules + AI scoring with reasons, hand-off to staff	Voice agent on the same workflow
Commercial	Catalogue, quote versions, accept / request changes, discount approval, Razorpay links, payment ledger, GST invoices in the business's own template, credit notes, refunds with owner approval	Line-item partial approval, automatic refund rules, e-invoicing (IRN)
Scheduling	Platform calendar + Google Calendar (business and staff), free/busy, slot hold, auto-assignment, reassignment, leave	Route planning
Integrations	Google Sheets (Leads + Payments tabs), Meta Conversions API, Meta Marketing API, email	More connectors
Reliability	Action statuses, retries, failed-actions screen, alerts, integration health	Advanced monitoring
Dashboards	Business, staff, Northstar admin, lead score dashboard, customer timeline, reports	Cross-tenant benchmarking

Explicitly out of MVP: voice calling, mass broadcast campaigns, accounting ledger, payroll/HR, inventory, custom mobile apps.

3. Users, roles and permissions

Three roles: the business admin controls everything in their workspace, staff see only their own work, and the Northstar admin runs the platform.

Capability	Business admin	Staff	Northstar admin
Leads	All leads	Assigned leads only	Support view only
Calendar	All staff, combined view	Own calendar only	—
Conversations	All, take over any	Assigned only	—
Quotes, payments, invoices	Full control, approvals	View for own jobs	—
Team	Invite, change role, deactivate, mark leave	—	—
Settings, workflow, integrations	Full control	—	—
Tenants, usage, failed jobs, pause	—	—	Full control

Staff invites

1. Admin enters the staff email and role under Team.
2. Staff receives an invite email (expires after 7 days), sets a password, logs in.
3. On first login, staff connects their own Google Calendar.
4. Admin can resend, change role or deactivate at any time; deactivation removes access immediately and keeps past records.

Privacy rule: personal Google Calendar events show to the admin as "Busy" only, without title or details. Platform-created jobs show in full.

4. Lead sources and attribution

WhatsApp is the only entry point: every channel carries a WhatsApp chat button, and the platform tags each lead with where it came from.

Source	How it reaches WhatsApp	How the source is captured
Meta click-to-WhatsApp ads	Ad opens a chat	Ad ID, campaign and click ID arrive with the first message (referral payload)
Website, email, Instagram/Facebook bio, other	Tracked chat link `northstar.com/c/<code>` or its QR code	Link logs the click (source, UTM tags, page), then opens WhatsApp with a pre-filled message carrying a short reference code; the platform matches the code on arrival
Direct message, no link	Customer messages the number	Tagged "Direct / Organic"

Chat Links section (dashboard)

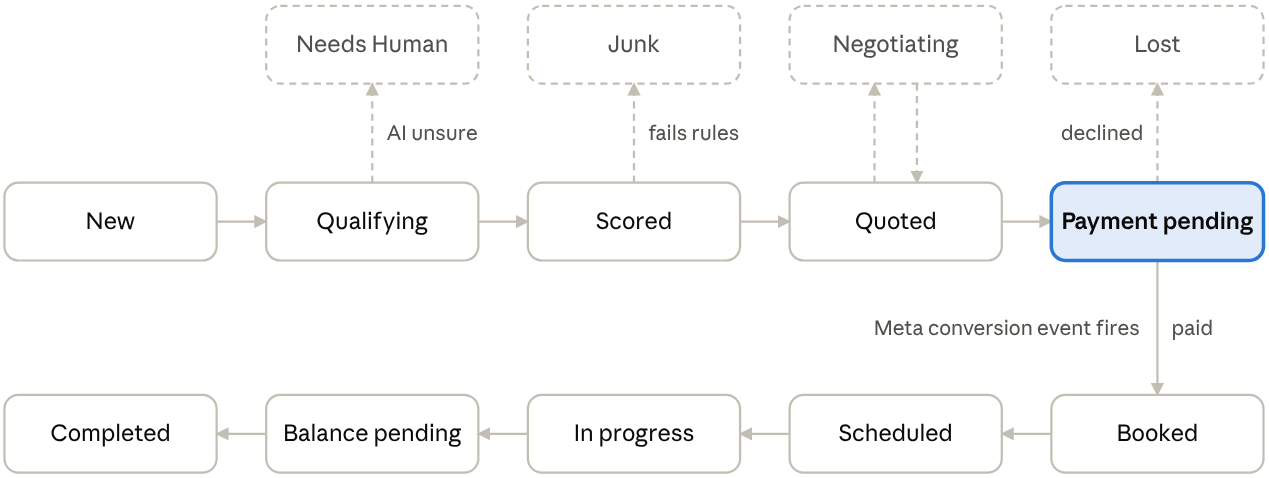
- Create, name and archive links; download QR codes.
- Per link: clicks, leads, average score, conversions, revenue.

Duplicate check: a new message from a phone number that already exists in the workspace attaches to the existing contact; the new source is added to the contact's source history.

5. End-to-end automated workflow

A lead moves through ten stages from first message to closed, with staff availability checked before the quote and the Meta conversion event fired when the payment link goes out.

Every lead runs ten stages from first message to completed



lead workflow · 10 stages, 4 side stages

Dashed boxes are side stages a lead can drop into; staff availability is checked at Quoted, and the Meta pre-final event fires when the payment link goes out.

#	Stage	What happens	Done by
1	New	Lead messages on WhatsApp; source captured; duplicate check by phone	System
2	Qualifying	AI asks the business's qualifying questions; collects service, date, time, location, budget, requirements	AI
3	Scored	Filter rules first (fail → Junk); AI score 0–100 with reasons → Hot / Warm / Cold	Rules + AI
4	Quoted	Free staff checked for the requested slot; only open slots offered; quote PDF from catalogue sent on WhatsApp	System (staff approve in approval mode)
5	Payment pending	Client accepts; Razorpay link (advance or full) sent; slot held; **Meta pre-final conversion event fires**	System

#	Stage	What happens	Done by
6	Booked	Razorpay confirms payment; GST invoice sent; free staff auto-assigned	System
7	Scheduled	Event created in platform calendar and Google Calendar (business + staff); reminders to client and staff	System
8	In progress / done	Staff marks the task complete from dashboard or WhatsApp	Staff
9	Balance pending	If the first payment was partial, balance link sent; reminders until paid	System
10	Completed	Balance paid; final invoice, thank-you and review request sent; Purchase event with value sent to Meta	System

Side stages: Needs Human (AI unsure, negotiation, complaint) · Negotiating (client asked for changes) · Junk (failed filter rules) · Lost (declined or silent after follow-ups) · Cancelled.

Branches

- No reply → automated follow-ups (default maximum 2) → task for staff.
- Request changes on quote → Negotiating → staff revises → new quote version.
- Payment link expires → slot released → one reminder with a fresh link.
- Payment after slot release → availability re-checked → confirm, or alert staff to reschedule.
- Balance unpaid after work → scheduled reminders → escalation to staff; delivery hold optional per business.

Approval mode (per business toggle)

- **Approval mode (default for new clients):** AI drafts quotes, payment links, bookings and non-template messages; staff tap Approve.
- **Automatic mode:** the same actions go out without approval, still within the rules (price limits, discount limits).

- Business admin switches modes in Settings; every action records which mode it ran under.

6. Configurable workflow engine

One engine runs every business; each business picks a template and switches steps on or off, so a new industry needs configuration, not new code.

Building blocks

- **Step types:** qualify, score, discovery call, site visit, counselling, quote, approval, advance payment, full payment, schedule, assign staff, service/job, batch assignment, delivery, balance payment, review request, close.
- **Per stage:** name, order, required fields, allowed next stages, automation triggers, message templates, owner/queue, timers, approval required (yes/no).
- **Simple conditions:** rules on lead fields, e.g. "budget > ₹50,000 → add site visit", "repeat customer → skip advance".
- **Versions:** a published workflow is locked; leads in progress keep the version they started on unless an admin migrates them.

Industry templates

Template	Flow
Shoot production (Reelrr)	The 10-stage flow in section 5
Agency	Enquiry → Discovery Call → Proposal → Retainer → Onboarding
AC / home service	Enquiry → Site Visit → Quote → Advance → Service
Coaching	Enquiry → Counselling → Admission → Fee Payment → Batch Assignment
Blank	Build from scratch

MVP boundary: a form-based editor (pick steps, set order, set conditions). The drag-and-drop visual builder is Phase 2.

7. AI agent and lead scoring

The AI talks, extracts and scores, but it only proposes actions; the workflow engine checks rules and permissions before anything is executed.

What the AI does

- Holds the qualifying conversation in the business's tone and language.
- Extracts structured fields (service, date, time, location, budget, requirements) with a confidence value and the message it came from.
- Answers only from the business's approved FAQ and catalogue.
- Hands off to staff when unsure, when the client negotiates or complains, or on low confidence.

Allowed tools (typed, validated, logged): `get_lead`, `update_allowed_fields`, `propose_stage_change`, `check_availability`, `draft_quote`, `create_callback_task`, `escalate_to_human`.

Never allowed: access another business's data, run database queries, mark payments paid, issue refunds, confirm bookings or send quotes outside the rules, follow instructions inside customer messages that try to change its behaviour.

Lead scoring

1. **Filter rules (set by the business):** location zone, minimum budget, service type, spam keywords. A failed rule → Junk (kept, not deleted, so staff can review).
2. **AI score (0–100)** on leads that pass, with stated reasons, e.g. "budget matches, date available, location outside core zone".
3. **Bands:** Hot, Warm, Cold, Junk; thresholds set per business.
4. **Score history** stored whenever the score changes during the conversation.

Lead score dashboard: distribution by band, average score by source and by Meta campaign/ad, conversion rate by band, hot leads not yet contacted.

Kill switch: AI can be paused per conversation (staff takeover), per business and platform-wide.

8. Commercial module

Every price the client accepts is tied to an exact quote version, every rupee is a ledger entry, and invoices follow GST rules, including credit notes for cancellations.

Quotes

- Built from the business's service catalogue as line items (price, HSN/SAC, tax rate).
- Every revision is a new version (v1, v2, v3); old versions stay readable.
- Client sees **Accept** / **Request Changes** buttons on WhatsApp. Accept links to that exact version; Request Changes moves the lead to Negotiating and hands it to staff.
- Discounts above a business-set limit need admin approval before the revised quote goes out.
- Partial approval (accept some line items) is Phase 2 unless pilot clients need it.

Payment ledger

- Every payment is its own record: type (full, advance, partial, balance) and status (pending, paid, failed, expired, refunded).
- Amount due is always calculated from quote total minus paid amounts, never typed.
- Razorpay Payment Links created per payment; paid status comes only from Razorpay's verified webhook, never from AI or a manual guess.
- Failed payment → fresh link sent; due-date reminders on a schedule.

Invoices

Item	Rule
Lifecycle	Draft → Issued → Partially Paid → Paid → Overdue → Cancelled
Cancellation	An issued invoice is never deleted; it is cancelled with a credit note
Numbering	Per business, per financial year, no gaps
Tax	CGST + SGST when client is in the same state, IGST when different; HSN/SAC and rate per service
Business details	GSTIN, legal name, billing address, state
Template	Business uploads its own Word (.docx) design with placeholders (e.g. `{{client_name}}`, `{{gst_amount}}`); field-mapping screen + preview. Default GST template for businesses without one. PDF-only businesses: Northstar team converts during onboarding

Item	Rule
Delivery	PDF sent on WhatsApp and stored in the customer timeline

Tax rules must be reviewed by a chartered accountant before launch. Mandatory e-invoicing (IRN) for larger businesses is Phase 2.

Refunds

- Cancellation policy set per business (e.g. full refund 48+ hours before, 50% within 48 hours, none same day).
- Refund amount calculated from the policy; owner approves; processed via Razorpay Refunds API; recorded in the ledger with a credit note.

9. Scheduling and calendar

The platform calendar is the master record, mirrored to Google Calendar; availability is checked before quoting, so a client never pays for a slot nobody can serve.

Availability and booking

1. At the quote stage, the system reads free/busy from every eligible staff member's Google Calendar plus platform jobs and leave.
2. Only open slots are offered to the client.
3. When the payment link goes out, the slot is held until the link expires.
4. On payment, free staff are auto-assigned (by skill and workload), and the event is created in the platform calendar, the business Google Calendar and the staff member's Google Calendar.

Views

- **Staff:** own calendar only (assigned jobs + own Google events).
- **Admin:** all staff in one view, colour-coded per person, filter by staff, day / week / month, side-by-side team view.

Manual reassignment (admin)

- From the appointment or by drag-and-drop in team view; free staff shown as available, busy staff greyed out but selectable with a warning.
- Both staff notified on WhatsApp and email; event moves between their Google Calendars.

- Client details and reminders unchanged; optional message to the client naming the new staff member.
- Logged in the appointment history (who, when, from whom to whom).

Leave: admin marks a staff member on leave for a date range; they are excluded from auto-assignment, and their existing appointments in that period are flagged for reassignment.

Cancel / reschedule: cancel releases the slot, removes calendar events, notifies staff and starts the refund flow; reschedule offers new open slots and moves calendars, reminders and assignment.

10. Integrations

Each business connects its own WhatsApp number, Razorpay account, Google account and Meta ad account; Northstar holds the platform-level apps and credentials that make those connections.

Integration	API	Used for	Connected by
WhatsApp	Official WhatsApp Cloud API (Northstar as Meta Tech Provider, Embedded Signup)	Inbound/outbound messages, templates, PDFs, delivery status	Business, in onboarding
Payments	Razorpay Payment Links, Refunds, webhooks	Advance/balance links, paid confirmation, refunds	Business
Calendar	Google Calendar API (OAuth)	Free/busy, create/update/delete events	Business + each staff member
Sheets	Google Sheets API (OAuth)	Live Leads and Payments tabs	Business
Meta outbound	Conversions API (business messaging)	Stage events matched on click ID	Business (ad account / dataset)
Meta inbound	Marketing API	Campaign names, ad names, spend for cost-per-lead reports	Business

Integration	API	Used for	Connected by
Email	Resend	Invites, verification, notifications	Northstar
AI	LLM API with tool calling (provider-agnostic)	Conversation, extraction, scoring	Northstar

Google Sheets sync

- Database is the source of truth; Sheets is a live mirror.
- Two tabs: **Leads** (Lead ID, name, phone, source, campaign, ad, chat link, score, band, score reason, stage, assigned staff, service, date/time, location, quote amount, created, updated, sync status) and **Payments** (Payment ID, Lead ID, name, invoice no., type, amount, GST, total, Razorpay link ID, Razorpay payment ID, status, paid at).
- Rows matched by platform ID, never row number; updates batched to stay within Google's per-project quota.
- Platform → sheet by default; selected fields (e.g. notes, stage) can be opted in to flow back. Conflicting edits on protected fields: platform wins, change goes to a review list.
- No message transcripts or tokens copied into sheets.

Meta Conversions API events

Event	Fires when	Purpose
Lead / qualified	Lead scored Warm or Hot	Light signal
Pre-final conversion	Payment link sent (stage configurable per business)	Main optimisation event
Purchase (with value)	Lead completed	Value signal for ROAS

Connection health: every integration shows connected / error / reconnect and last successful sync.

11. Dashboards

Three dashboards from one React app, each showing only what that role is allowed to see.

Business dashboard (admin)

Section	Contents
Overview	Today's leads, hot leads, revenue collected, pending payments, upcoming jobs, items needing attention
Leads (CRM)	All leads with source, score, stage, assigned staff; filters; open a lead → customer timeline
Customer timeline	One chronological record: messages (AI and human), staff-logged calls, quote versions, payments, refunds, invoices, appointments, task changes, approvals, delivery evidence; each entry shows who (AI, system, staff) and when
Lead score	Distribution by band, average by source and campaign/ad, conversion by band, hot leads not contacted
Conversations	WhatsApp inbox, Needs Human queue, take over / hand back to AI
Calendar	All staff, team view, reassignment, leave
Quotes, invoices, payments	Versions, ledger, invoice PDFs, credit notes, refunds awaiting approval
Team	Staff, roles, workload, invites
Chat Links	Links, QR codes, clicks → leads → conversions per link
Reports	Funnel conversion, source/campaign performance with cost per qualified lead, time per stage, revenue
Failed actions	Failed or Needs Review actions, error, retry / skip
Integrations	Connection status, last sync, reconnect
Settings	Workflow, approval mode, rules, catalogue, invoice template, GST details, cancellation policy

Staff dashboard: my tasks, my calendar, my assigned leads and conversations; mark task done; add notes and log calls.

Northstar admin panel: all businesses with status and subscription, message and AI usage per business, failed jobs and webhook errors, integration health across tenants, pause a business.

12. Client onboarding wizard

A new business goes live through nine guided steps, ending with a test lead run end to end; the workspace activates only after payment or an explicit trial override.

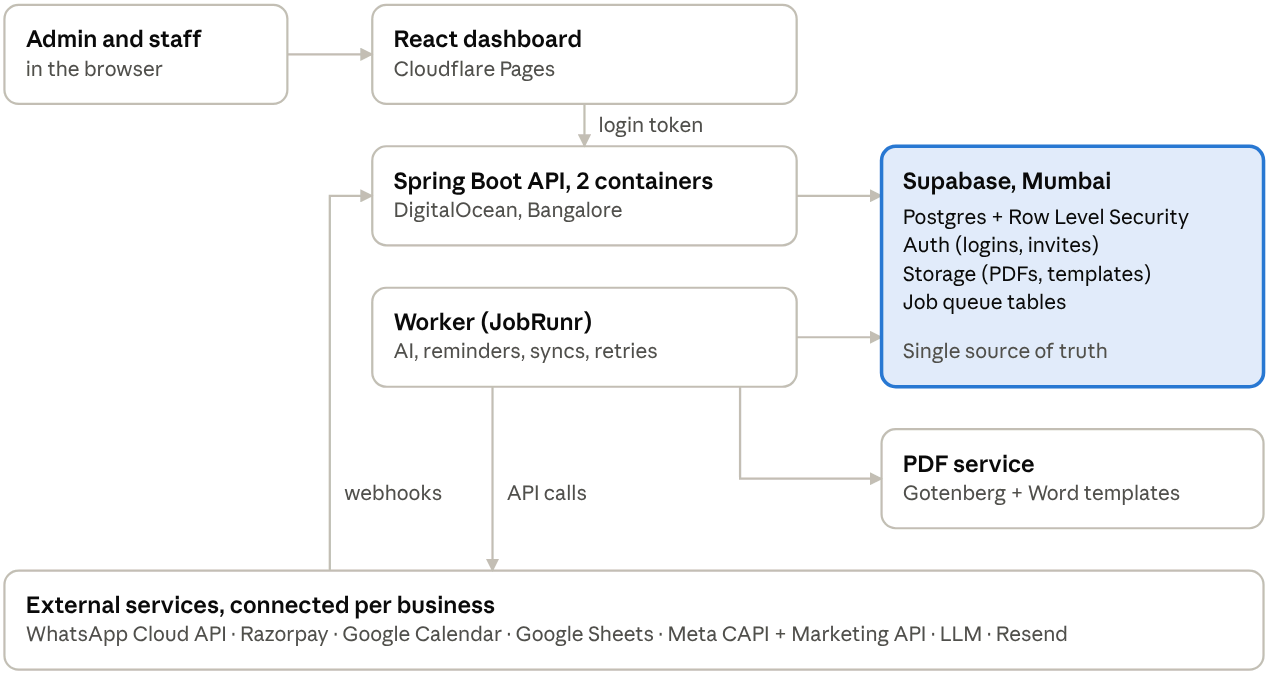
1. **Sign up and verify** email.
2. **Create workspace:** business name, industry, timezone, working hours, language, brand tone.
3. **Business details:** service catalogue with prices, HSN/SAC and tax rates; GSTIN, legal name, address, state.
4. **Invoice template:** upload Word template or choose the default; map fields; preview.
5. **Team:** invite staff by email with roles and skills; staff connect Google Calendars.
6. **Integrations:** WhatsApp (required), Razorpay (required), Google Calendar, Google Sheets, Meta ad account.
7. **Workflow settings:** template, qualifying questions, filter rules, score thresholds, advance %, reminder timings, cancellation policy, Meta conversion stage, approval mode.
8. **Chat links:** generate tracked links and QR codes per source.
9. **Test run:** a dummy lead goes through the full flow; check messages, CRM record, sheet row, calendar event, invoice; go live.

Readiness checklist shown per business: each step green / pending, with what is blocking. WhatsApp number verification can add external lead time, so activation is never promised as instant.

13. Architecture and tech stack

React frontend, Spring Boot modular monolith, Supabase Postgres in Mumbai as the single source of truth, and background jobs for everything slow or external.

One backend, one database; background jobs make every external call



system architecture · frontend, backend, database, external services

Webhooks come into the API, which stores the event and queues a job; the worker does every slow or external call, so an outage at Google or Meta never blocks the dashboard.

Layer	Tool	Role
Frontend	React + TypeScript (Vite), Tailwind CSS, shadcn/ui	Business, staff and Northstar admin dashboards; onboarding
Calendar UI	FullCalendar	Day/week/month, team view, drag-and-drop
Charts	Recharts	Score dashboard, funnels, reports
Frontend validation	Zod	Forms
Backend	Spring Boot (Java 21), modular monolith	APIs, workflow engine, rules, integrations, webhooks
Backend validation	Jakarta Bean Validation	Webhooks, API input, AI output

Layer	Tool	Role
Database	Supabase PostgreSQL (Mumbai), Flyway migrations	Source of truth
Tenant isolation	Spring Security tenant filter + Supabase Row Level Security	Two layers
Auth	Supabase Auth, tokens verified by Spring Security	Login, verification, invites, roles
Storage	Supabase Storage, signed short-lived links	Quotes, invoices, templates, delivery files
Background jobs	JobRunr on Postgres	Reminders, follow-ups, retries, syncs, link expiry, slot holds
AI	Spring AI + LLM with tool calling	Provider-agnostic
Documents	poi-tl (Word templates) + Gotenberg (PDF)	Quotes and GST invoices
QR codes	ZXing	Chat-link QR codes
Integrations	WhatsApp Cloud API, Razorpay Java SDK, Google API Java clients, Meta CAPI + Marketing API, Resend	See section 10
Frontend hosting	Cloudflare Pages (or Vercel)	Static React app
Backend hosting	DigitalOcean App Platform, Bangalore	API (2 containers), worker, PDF service, staging
Monitoring	Sentry	Frontend + backend errors
Code and CI/CD	GitHub + GitHub Actions	Reviews, tests, deploys
Testing	JUnit 5 + Testcontainers, Vitest, Playwright	Including cross-tenant access test

Backend modules (one deployable): tenant & access · contacts & conversations · workflow runtime · CRM & jobs · AI orchestration · commercial (quotes, ledger, invoices) · scheduling · integrations · reporting · admin. Modules call each other through service

interfaces only, so any can be split out later.

14. Data model

Every business-owned row carries `tenant_id`, enforced in application queries and by Row Level Security; UUID keys, UTC timestamps, typed columns for queryable fields and JSONB only for business-specific extras.

Table group	Key fields
tenants	id, name, slug, timezone, locale, business_profile, gstin, state, plan, status, approval_mode
users / memberships	auth_user_id, tenant_id, role (admin / staff), skills, status, invited_at, leave periods
contacts	tenant_id, normalized_phone (unique per tenant), name, consent_state, opt_out_at, sources history
conversations / messages	tenant_id, contact_id, channel, provider_message_id (unique), direction, type, status, timestamps
chat_links / link_clicks	tenant_id, code, source, utm fields, clicks, matched lead
workflows / workflow_versions / stages	tenant_id, template, version, config JSON, published_at (immutable once published)
leads	tenant_id, contact_id, workflow_version_id, stage, source, campaign, ad, click_id, requirements, score, band, score_reasons, assigned_staff_id
lead_score_history	tenant_id, lead_id, score, reasons, created_at
catalogue_items	tenant_id, name, price, hsn_sac, tax_rate
quotes / quote_versions / quote_lines	tenant_id, lead_id, version, lines, totals, status (draft, sent, accepted, changes requested, superseded)
payments (ledger)	tenant_id, lead_id, quote_version_id, type, amount, status, razorpay_link_id, razorpay_payment_id, paid_at

Table group	Key fields
invoices / credit_notes	tenant_id, number (per FY), type, lines, CGST/SGST/IGST, status, pdf_path
invoice_templates	tenant_id, docx_path, field_mapping, is_default
refunds	tenant_id, payment_id, amount, policy_applied, approved_by, razorpay_refund_id, status
appointments / slot_holds	tenant_id, lead_id, staff_id, start, end, google_event_ids, hold_expires_at, status
tasks	tenant_id, lead_id, owner, due_at, status
actions (action log)	tenant_id, lead_id, type, status (pending, processing, succeeded, failed, needs review), attempts, error, idempotency_key, mode (approval / automatic)
timeline_events	tenant_id, lead_id, actor (AI / system / staff id), event type, reference, created_at (append-only)
integrations / sync_mappings	tenant_id, provider, encrypted credentials reference, scopes, sheet/tab ids, field mapping, last_sync, status
meta_events	tenant_id, lead_id, event name, value, sent_at, response
audit_log / usage	tenant_id, actor, action, target, safe metadata; usage units per message, AI call, storage

Secrets and cross-tenant configuration never live in user-editable JSON.

15. Reliability, failure handling and data pipelines

The workflow never stops silently: every automated action has a status, failures retry automatically, and anything that still fails lands in front of a person.

Pipeline

1. **Inbound:** WhatsApp, Razorpay and Meta webhooks hit Spring Boot; signature verified; event ID checked for duplicates; event stored; job queued. The webhook replies fast and never does slow work inline.

2. **Processing:** a worker loads the lead, runs AI if needed, and the rules engine decides the next action based on stage, settings and approval mode.
3. **Outbound:** every external call (WhatsApp send, Razorpay link, Calendar event, Sheets row, Meta event, PDF) is an action in the job queue with an idempotency key.

Rules

- **Action status:** Pending → Processing → Succeeded / Failed / Needs Review.
- **Retries:** exponential backoff; after the retry limit → Failed, shown on the Failed Actions screen with the error, a Retry and a Skip button.
- **Alerts:** admin alerted when an action fails repeatedly (e.g. "Invoice for Rahul Sharma failed 3 times"); Northstar admin sees failures across all tenants.
- **Duplicate protection:** provider event IDs and idempotency keys mean a repeated webhook or job never creates a second lead, payment, invoice or message.
- **Outbox for Sheets and Meta:** changes written to an outbox table first, then synced in batches; last successful sync shown.
- **Degraded mode:** if Google Calendar or Sheets is down, the platform keeps working and catches up on reconnect.
- **Kill switches:** pause automation per conversation, per business, platform-wide.

Scaling rules (from day one): stateless Spring Boot containers, all heavy work through the queue, every table indexed by `tenant_id`, message history archived after a retention period.

16. Edge cases and expected behaviour

Each case below gets an automated test before launch; the behaviour is defined now, alongside the architecture, not left to the end of development.

Situation	Expected behaviour
Same lead webhook arrives twice	No duplicate contact, lead or message
Meta authorisation expires	Connection error visible; reconnect option; automation for that integration paused, not failed silently
Server fails while saving a lead	Event kept and retried; no lost lead

Situation	Expected behaviour
Phone number missing	Defined rule: WhatsApp leads always have one; manual entries rejected without a phone
Lead reassigned to another user	Ownership history preserved
User deletes a lead	Admin only; soft delete with audit entry; retention rules apply
Two users edit the same lead	Version check; second save warned, never silently overwrites
Workspace subscription ends	Defined grace period, then read-only; automation paused; data kept per retention policy
Client pays twice, or Razorpay webhook repeats	One payment record, one invoice; extra payment flagged for refund
Payment link expires unpaid	Slot released; one reminder with a fresh link
Client pays after the slot was released	Availability re-checked; confirm if free, else staff alerted to reschedule
Assigned staff becomes unavailable	Auto-reassign to free staff, else alert admin
Client wants to reschedule or cancel after paying	Hand-off to staff; refund per business policy with owner approval
Client replies STOP or opts out	All automated messages stop at once, scheduled ones cancelled
AI unsure or client negotiates	Needs Human; conversation flagged
Quote sent with a wrong price	Staff revises; new version sent; history kept
Google Calendar disconnected	Platform calendar keeps working; error shown; sync resumes on reconnect
Client messages after 24 hours of silence	Approved WhatsApp template used, per Meta's customer-service window rule
Balance unpaid after work	Scheduled reminders, then escalation; delivery hold optional
Invoice generation fails	Action Failed → retries → Failed Actions screen; payment record unaffected

Situation	Expected behaviour
Sheet edited by a user on a protected field	Platform value restored; change sent to review list

17. Security, privacy and compliance

Tenant isolation is a release gate: no build ships if a test shows one business can read or change another's data.

- **Tenant isolation:** tenant taken from the verified login token, never from the request; RLS on every tenant table; cross-tenant test on every pull request.
- **Auth:** secure sessions, MFA for admins where available, invite expiry, immediate offboarding, audited admin actions.
- **Secrets:** provider tokens encrypted at rest; service keys on the server only; separate staging and production; credential rotation.
- **Webhooks:** signature verification, payload size limits, duplicate rejection, minimal stored data.
- **AI privacy:** only necessary context sent to the model; provider data-retention settings configured; prompts and system instructions never exposed to customers.
- **WhatsApp policy:** approved templates where required, customer-service window honoured, opt-in recorded, opt-out immediate, no unsolicited bulk messaging, quality rating monitored.
- **Payments:** card data never stored; generated invoices never marked paid without Razorpay confirmation.
- **Data residency:** database and storage in Mumbai, backend in Bangalore; supports India's DPDP Act obligations.
- **Data rights:** documented retention, export and deletion workflows; backup expiry defined; message transcript retention minimised.
- **Incident readiness:** audit trail, backup and restore drill, error alerting, credential revocation procedure, support contact.

Before commercial launch, privacy notices, tax treatment, messaging consent and payment terms are reviewed by Indian legal and tax advisers.

18. Infrastructure cost and capacity

The launch setup costs about $245 a month (≈ ₹23,640), about ₹2.84 lakh for year 1 or ₹3.26 lakh with a 15% buffer, and serves 50–100 client businesses.

Item	Service	Monthly (USD)
Database, auth, storage	Supabase Pro + Small compute + staging	60
React frontend	Cloudflare Pages	0
Spring Boot API (2 × 1 vCPU / 2 GB)	DigitalOcean App Platform	50
Background worker (JobRunr)	DigitalOcean App Platform	25
PDF service (Gotenberg)	DigitalOcean App Platform	25
Staging backend	DigitalOcean App Platform	25
Email	Resend	20
Error tracking	Sentry	26
Code and CI/CD (3 users)	GitHub Team	12
Domain	—	2
Total		**245**

Converted at ₹96.5 per USD; planning estimates to confirm on provider pricing pages. Not included: ~18% GST on provider bills (usually claimable as input credit), WhatsApp message charges, LLM usage, Razorpay fees, team costs.

Capacity tiers

Setup	Client businesses	Approx. monthly infra
Launch (above)	50–100	≈ ₹27k
Growth: bigger Supabase compute, 3–4 API containers, 2 workers	250–500	≈ ₹55–80k

Setup	Client businesses	Approx. monthly infra
Scale: read replica, dedicated worker pool, message archiving	1,000+	≈ ₹1.5L+

Growth limits to watch: Google Sheets API quota shared across all tenants (batch writes, request higher quota before ~100 clients); LLM provider rate tier; database storage from message history; onboarding and support capacity.

19. Build phases with Claude Code

The build runs in ten phases, each closed by a gate that must pass before the next starts; external approvals (Meta, Google) start in week 1 because they usually set the real timeline.

Step 0, week 1, in parallel

- [ ] Meta: business verification, Meta app, Tech Provider application, WhatsApp / ads / Conversions API permissions
- [ ] Google Cloud: project, OAuth consent screen, Calendar + Sheets APIs, app verification submitted
- [ ] Razorpay: account and KYC; decide how clients connect their accounts
- [ ] Supabase: staging + production projects in Mumbai
- [ ] DigitalOcean (Bangalore), domain, Resend domain, Sentry, LLM API key, GitHub organisation

Repository: `northstar/` with `CLAUDE.md`, `docs/` (this document and specs), `frontend/`, `backend/`, `db/migrations/`, `.github/workflows/`.

Working method per module: plan in Claude Code's plan mode → review the plan → implement in small pieces → tests pass → developer review (always for payments, auth, tenant data, webhooks) → pull request → CI → merge. At least one experienced Java developer supervises.

Phase	What gets built	Gate
1. Foundation	Repo, CI, deploy pipeline; schema, migrations, RLS; Supabase login verified by Spring Security; tenants, roles, staff invites; dashboard shell	Two test businesses; automated test proves neither sees the other's data
2. WhatsApp + inbox	Webhook (signature, duplicates), message store, send, templates, delivery status; inbox with takeover; opt-out	Real message in, reply out; duplicate webhook creates nothing
3. Workflow engine	Stages, step types, templates, versions; action log + statuses; JobRunr jobs, retries, failed-actions screen, alerts; approval mode	Test lead runs the Reelrr template end to end; forced failure appears on failed-actions screen
4. AI + scoring	Spring AI tools, qualifying conversation, extraction; filter rules + AI score; hand-off; score dashboard	AI qualifies and scores with reasons, hands off when unsure, cannot act outside its tools
5. Commercial	Catalogue, quote versions, accept / changes, discount approval; template upload + PDF; Razorpay links + webhooks; ledger; GST invoices, credit notes; refunds	Quote → pay advance → invoice → balance → final invoice in Razorpay test mode
6. Scheduling	Google Calendar per staff, free/busy, slot hold, auto-assign; calendar views; reassignment, leave, cancel, reschedule	Only open slots offered; event lands in the right calendars; reassignment moves it
7. Outbound integrations	Sheets sync; Meta CAPI + Marketing API; chat links + QR	Sheets update in seconds; events visible in Meta Events Manager; link sources tagged
8. Dashboards + admin	Overview, CRM, timeline, reports, integration health; onboarding wizard; Northstar admin	New business goes live through the wizard unaided

Phase	What gets built	Gate
9. Hardening	All edge-case tests, load test, security review, backup/restore drill, alerts, production deploy	Every edge case in section 16 has a passing test; restore works
10. Pilot	Reelrr + 2–4 design partners on real numbers	Real leads flow end to end; weekly fix cycle

Phases 1–3 run in order; 4–7 can overlap with more than one developer.

MVP acceptance criteria

- [ ] Two or more businesses operate independently; no cross-tenant access through UI, API, jobs, exports or callbacks
- [ ] Inbound WhatsApp message creates or updates one contact and lead; staff can pause and take over
- [ ] Reelrr workflow runs all ten stages, including partial payment and balance
- [ ] Workflow config can disable payment or booking steps without code changes
- [ ] AI tool calls are validated, tenant-scoped, idempotent and audited; uncertain cases go to staff
- [ ] Sheets sync upserts by ID, retries visibly, handles conflicts and disconnects
- [ ] Opt-out stops all automated messages; WhatsApp window and template rules enforced
- [ ] Meta events fire at the configured stages and match on click ID
- [ ] Northstar admin sees tenants, failed jobs, webhook errors, usage, and can pause a business

20. Open decisions

These need an answer before or during Phase 0–1; none blocks starting the foundation.

- [ ] Who pays WhatsApp message charges: billed to each client's own Meta account, or bundled into the Northstar plan?
- [ ] How clients connect Razorpay: their own API keys, or a Razorpay partner integration?
- [ ] Which LLM provider to start with, and the per-tenant AI usage allowance
- [ ] Lead score thresholds for Hot / Warm / Cold defaults
- [ ] Subscription end behaviour: grace period length and read-only rules

- [ ] Data retention periods for messages, files and audit logs
- [ ] Default cancellation and refund policy offered in templates
- [ ] Chartered accountant review of GST, invoice numbering and credit-note rules
- [ ] Whether line-item partial quote approval is needed for pilot clients